IMPLEMENTATION · SDLC & COMMERCIAL PROPOSAL

JPMC AI-Assisted Connected Patient Flow System

A pilot-first proposal covering connected OPD and Emergency workflows, SDLC delivery planning, clinical-safety governance and an indicative commercial estimate for a measured 24–28 week implementation.

PREPARED FOR

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Karachi, Pakistan · Indicative planning and pilot proposal

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Document Control

Item	Detail
Project	AI-Assisted Connected Patient Flow System
Primary scope	One OPD + Emergency Department pilot, followed by controlled production rollout
Delivery approach	SDLC: Planning, Analysis, Design, Development, Testing, Deployment & Maintenance
Commercial status	Indicative estimate; subject to discovery, site survey, procurement and integration assessment
Clinical safety principle	AI assists by organizing, predicting and flagging; clinicians make diagnosis, triage and prescribing decisions.

Executive Summary

JPMC requires a connected digital workflow that reduces administrative friction in OPD and emergency care while giving clinicians timely access to relevant patient information. The proposed system provides a patient mobile/web portal, doctor and administration dashboards, digital registration and tokening, live queue and doctor-availability visibility, digital prescriptions/tests, payment/discharge workflows, and controlled AI assistance.

The first delivery is a measured pilot in one OPD and one Emergency Department workflow. The pilot establishes baseline operational data, validates interfaces with JPMC stakeholders, proves reliability and staff adoption, and creates an evidence-based decision for hospital-wide scaling. The implementation target is 24–28 weeks (approximately 6–7 months), excluding delays caused by third-party integrations, approvals, procurement or infrastructure readiness.

1. Objectives and Problem Statement

1.1 Current operational problems

- Manual or fragmented registration and appointment processes can create avoidable queues and repeated data entry.
- Doctors may spend consultation time locating history, reports, medications, allergies and current vitals across disconnected records.
- Patients often lack real-time visibility of token status, queue position, expected wait and clinician availability.
- Emergency arrivals can face administrative steps before clinical prioritization if workflows are not explicitly designed for parallel processing.

- Hospital leaders need timely operational information on volume, bottlenecks, utilization, queue performance and system exceptions.

1.2 Project objectives

Objective	Measurable pilot indicator
Reduce administrative waiting	Registration and token issue time; average queue wait versus baseline
Improve clinician readiness	Percentage of encounters with accessible structured history, reports and vitals
Enable safe emergency fast track	Arrival-to-triage and triage-to-treatment timestamps by priority category
Improve patient visibility	Live queue availability, notification delivery and patient satisfaction
Strengthen governance	Role-based access compliance, audit-log completeness and security-test results
Improve operations	Dashboard use, doctor availability accuracy, system uptime and support ticket trends

2. Scope and Solution Architecture

2.1 Functional scope

Module	Pilot capabilities
Patient app / web portal	Account/login, appointment request, digital registration, symptom/history entry, report access/upload, token, queue status, waiting estimate, prescription/test visibility, payment/discharge notices and follow-up reminders
Doctor dashboard	Secure sign-in, availability status, assigned queue, patient history, reports, medication/allergy view, vitals, AI-organized summary, consultation notes, digital prescription and test orders
Nurse / triage station	Quick identification, basic information capture, manual vital entry or device capture, nurse-led triage priority, emergency routing and escalation
Administration dashboard	Queue and workload view, registration performance, doctor availability, alerts, user/role administration, operational reports and audit search
Integration layer	Controlled interfaces to existing HIS, billing, pharmacy, laboratory, radiology, payment services and notifications where technically feasible

2.2 AI functions and boundaries

AI function	Inputs	Output	Human control
Patient information organizer	Complaint, duration, conditions, medicines, allergies, uploaded reports	Structured encounter summary	Doctor validates before use

AI function	Inputs	Output	Human control
Waiting-time prediction	Queue length, provider availability, service time, department and emergency load	Estimated waiting range	Staff manage queue and exceptions
Queue/routing assistance	Policy rules, capacity and queue state	Alternative queue/provider suggestion	Hospital staff decide
Vitals flagging	BP, pulse, temperature, SpO_2, weight and configured thresholds	Review flag for unusual readings	Nurse/doctor assesses clinically

The system shall not autonomously diagnose, prescribe, determine emergency severity, or overrule clinician judgment. All predictions and summaries must show their source information, timestamp, confidence/limitations where applicable, and a clear clinician review workflow.

2.3 Technical architecture

- Presentation layer: Android/iOS patient application (or responsive web first), patient web portal, doctor/nurse/admin web dashboards, and queue-display interface.
- Application layer: API gateway, identity and access service, workflow/queue service, appointment service, encounter/document service, notification service, billing/payment adapters and reporting service.
- Data layer: transactional relational database, encrypted document/object storage for reports, audit-log store, analytics repository and backup repository.
- AI layer: secured summarization service, feature pipeline for wait-time model, rules engine for vital flags and monitored model-serving endpoint.
- Integration layer: API adapters and message queues. Health information exchange should map to HL7 FHIR resources where JPMC systems and project scope support it.
- Infrastructure: segregated development, test/staging and production environments; monitoring, centralized logging, backup, disaster recovery and controlled release pipeline.

FHIR is a widely used API-focused standard for representing and exchanging clinical and administrative health information; its modular resources make it suitable as an interoperability mapping target, while local systems may still require adapters and data-mapping work.

2.4 Installation setup

Setup area	Implementation requirement
Site survey	Map OPD/ER workflow, network coverage, power backup, workstation locations, privacy constraints, devices and existing HIS interfaces
Network	Secure wired/Wi-Fi connectivity, VLAN/segmentation where approved, firewall rules, monitored bandwidth and failover plan
Identity	Named user accounts, MFA where feasible, RBAC, password policy, account approval/deactivation procedure

Setup area	Implementation requirement
Endpoints	Registered tablets/workstations, hardened browsers, automatic session lock, endpoint inventory and support ownership
Hardware	Biometric devices, queue displays, kiosks, BP/pulse/SpO_2/temperature/weight devices only after approved device list and integration validation
Hosting	JPMC-approved on-premise/private-cloud environment, encryption keys, backup retention, recovery targets and production access controls
Operations room	Support contacts, escalation matrix, downtime procedure, helpdesk channel, change-control process and training materials

3. SDLC Delivery Plan

3.1 Planning — Weeks 1–3

Activity	Technical detail / deliverable	Owner
Kick-off and governance	Project charter, steering committee, decision rights, risk register and communication plan	PM + JPMC sponsor
Workflow observation	Map current OPD/ER patient journey, bottlenecks, handoffs, exceptions and baseline measures	Business analyst + clinical leads
Site/integration survey	Inventory HIS, billing, pharmacy, lab, radiology, identity, devices, networks and available APIs	Architect + JPMC IT
Pilot definition	Select one OPD and ER workflow; define users, daily volume, language needs, devices and acceptance criteria	Steering committee
Privacy/security planning	Data classification, consent/notice needs, RBAC matrix, audit events, retention, hosting approval and threat model	Security lead + JPMC IT/legal

3.2 Analysis — Weeks 2–5

Activity	Technical detail / deliverable
Requirements	User stories, use cases, non-functional requirements, exception paths, service-level assumptions
Data analysis	Data dictionary, source-of-truth decision, patient identity matching rules, data-quality assessment and migration approach
Integration analysis	Interface specifications, authentication, field mapping, error/retry handling, test data and ownership
AI analysis	Permitted inputs, output format, clinician validation, evaluation dataset, bias/safety review and fallback when AI is unavailable
Baseline study	Measure current registration time, queue wait, triage-to-treatment time, utilization and satisfaction before pilot

3.3 Design — Weeks 4–7

Design work	Technical detail / deliverable
UX/UI	Clickable prototypes for patient, clinician, triage, admin and queue-display journeys; usability review with hospital users
Solution design	Architecture diagram, component boundaries, API specifications, database schema, event flows and deployment topology
Security design	RBAC matrix, authentication flow, encryption design, audit schema, privileged-access rules, backup and recovery design
Clinical/workflow design	Triage workflow, escalation logic, documentation templates, digital prescription/test workflow and downtime forms
Test design	Traceability matrix, functional/security/performance/UAT test cases, test-data plan and defect workflow

3.4 Development — Weeks 8–15

- Build core backend APIs, patient identity/profile, registration, appointment/token, queue, notifications and document services.
- Build patient portal/app and clinician, nurse/triage and administration dashboards against approved prototypes.
- Build role-based permissions, audit logging, encrypted document storage and operational reports.
- Build integration adapters iteratively using sandbox/test environments before production connectivity.
- Build AI summary, waiting-time estimate and vitals flagging behind feature flags; ensure manual workflow remains usable if AI is disabled.
- Use source control, peer review, automated builds, dependency scanning, configuration management and environment-specific secrets.

3.5 Testing — Weeks 16–19

Testing is a release gate, not a final activity. Defects are logged, assigned, corrected, retested and closed only after evidence is attached.

Test type	Examples / exit requirement
Unit and API testing	Core business rules, validation, authentication, duplicate prevention and integration error handling
System/integration testing	End-to-end registration-to-consultation-to-prescription/payment/discharge paths and external interfaces
Security testing	RBAC verification, session control, input validation, dependency review, audit events and vulnerability remediation

Test type	Examples / exit requirement
Performance testing	Expected pilot concurrency, queue updates, dashboard response, report upload and recovery behavior
Usability/accessibility testing	Staff and patient tasks can be completed within agreed workflow and language/accessibility needs
AI evaluation	Summary factuality against source record, wait prediction error, false alerts, clinician acceptance and safe fallback
UAT	Named JPMC users complete approved scenarios; critical defects are zero and high defects are formally accepted or resolved

4. Deployment, Beta and Quality Assurance

4.1 Beta rollout — Weeks 20–23

Stage	Duration	Activities	Go/no-go evidence
Internal alpha	1 week	Developer/QA test in staging; simulated clinical flows and integration test data	No open critical defects; security baseline passed
Controlled beta	2 weeks	Limited named staff, selected hours/queues, parallel manual fallback and daily review	Workflow completion, support volume, data accuracy and clinical feedback acceptable
Pilot expansion	1 week	Broader OPD/ER use, monitoring, training reinforcement and operational dashboard review	Pilot KPIs stable; JPMC sponsor approval
Production go-live	Weeks 24–28	Controlled cutover, hypercare, on-site/remote support, daily issue triage	Acceptance criteria and operational readiness signed off

4.2 Production deployment checklist

- Approve production change window, rollback plan, downtime communications and named decision makers.
- Create production environment; apply hardening, firewall rules, TLS certificates, encrypted backups, monitoring and alerting.
- Provision named users and role assignments; verify access with representative patient, nurse, doctor, billing and admin scenarios.
- Validate device connectivity, kiosk/display placement, time synchronization, printer/document workflows and power/network contingencies.
- Migrate only approved pilot data; reconcile counts, test identity matching and document data-provenance decisions.
- Enable integrations one at a time with monitoring; retain manual fallback and reconciliation process.

- Train users by role; issue quick-reference guides and record attendance/competency confirmation.
- Run go-live readiness review and obtain written approval before enabling live patient use.

4.3 Quality assurance framework

QA control	How it will be applied
Requirements traceability	Every requirement maps to design, build item, test evidence and acceptance result
Clinical validation	Doctors/nurses validate screen content, safety wording, triage process and clinical workflow before release
Code quality	Peer review, static analysis, dependency checks, branch protection and automated build validation
Security QA	Threat modeling, RBAC tests and audit-log verification. OWASP ASVS-style controls guide web application security requirements and verification.
Data QA	Validation rules, duplicate detection, integration reconciliation, error queues and source/timestamp visibility
Release control	Versioned release notes, approval gates, rollback capability and post-release monitoring
Metrics review	Weekly pilot scorecard: availability, latency, defects, support tickets, workflow completion and KPI trends

4.4 Acceptance criteria

- All critical pilot workflows pass UAT with documented evidence.
- No unresolved severity-1 defects; severity-2 defects have approved mitigation and target dates.
- RBAC, audit logs, backups, recovery test and monitoring are verified.
- Approved pilot users complete training; helpdesk and escalation procedures are operational.
- AI outputs are clearly advisory, traceable and can be disabled without blocking core clinical workflow.
- JPMC signs the pilot acceptance report after agreed KPI and readiness review.

5. Team, Stakeholders and Job Descriptions

Role / stakeholder	Key job description and accountability
JPMC Executive Sponsor	Approves scope, funding, policy decisions and scale-up; resolves cross-department blockers.
JPMC Project Owner / Hospital Administrator	Owns hospital coordination, operational decisions, user availability, site readiness and acceptance.
Clinical Lead (Doctor)	Validates clinical workflow, clinician dashboard content, AI output appropriateness and patient-safety escalation.

Role / stakeholder	Key job description and accountability
Nursing / Triage Lead	Defines triage steps, vital capture process, emergency handoffs, staff training and safety feedback.
JPMC IT / HIS Lead	Provides system inventory, integration access, network/hosting readiness, security approval and operational handover.
Billing / Pharmacy / Lab Leads	Validate financial, medication, diagnostic and reconciliation workflows where included.
Data Protection / Legal / Compliance	Reviews privacy notice/consent, retention, access governance, vendor obligations and incident procedure.
Project Manager	Plans delivery, budget, risks, reporting, change control, stakeholder meetings and milestone acceptance.
Solution Architect	Owns end-to-end architecture, integration approach, deployment topology, non-functional requirements and technical decisions.
Business Analyst / Healthcare Consultant	Conducts discovery, maps workflows, writes requirements, supports UAT and maintains traceability.
Backend Developers (2)	Build APIs, data model, workflows, integration adapters, audit logging and business logic.
Mobile/Web Developers (2)	Build patient and staff user interfaces, accessibility/responsive behavior and API integration.
AI/ML Engineer	Builds and monitors wait-time model, feature pipeline, evaluation and drift/performance monitoring.
NLP/AI Engineer	Builds structured-summary workflow, prompt/model controls, evaluation and human-review mechanisms.
UI/UX Designer	Produces prototypes, design system, user testing sessions and implementation-ready interface specifications.
QA Engineers (2)	Create test plans, automate regression where practical, execute testing, manage defects and prepare release evidence.
DevOps/Security Engineer	Builds CI/CD, environments, monitoring, secrets management, hardening, backup/recovery and incident support.
Support / Training Lead	Creates role-specific training, quick guides, adoption support, helpdesk triage and knowledge transfer.

Governance cadence

Forum	Frequency	Purpose
Steering committee	Biweekly during delivery; monthly after go-live	Scope, risks, funding, cross-department decisions and stage approvals
Clinical safety review	Weekly during design/test/beta	Triage, AI boundaries, usability, incidents and clinical acceptance

Forum	Frequency	Purpose
Technical delivery meeting	Weekly	Architecture, integrations, security, release readiness and blockers
Daily beta/go-live huddle	Daily during beta and first 2 weeks after go-live	Defects, workflow disruptions, metrics, support and next-day action

6. Timeline and Cost Estimate

6.1 Indicative timeline

Phase	Duration	Calendar weeks	Primary outputs
Planning and discovery	2–3 weeks	1–3	Charter, site survey, pilot scope, baseline, risk/security plan
Analysis	2–3 weeks	2–5	Requirements, data/integration map, KPI definitions
Design	3 weeks	4–7	UX prototypes, architecture, security/test designs
Core development	7–8 weeks	8–15	Portals, dashboards, APIs, audit and workflow modules
AI/ML development	4–5 weeks, overlaps core build	10–15	Summary, wait estimate, flags and evaluation framework
Integration	3–4 weeks, overlaps	12–18	HIS/billing/pharmacy/lab adapters in test environment
Testing and security	2–3 weeks	16–19	System, performance, security and UAT evidence
Beta and pilot deployment	3–4 weeks	20–23	Controlled use, monitoring, feedback and refinements
Training and go-live	1–2 weeks	24–28	Production launch, hypercare, acceptance and handover

Total target duration: approximately 24–28 weeks. Timing is contingent on timely JPMC decisions, stakeholder availability, access to integration environments, approved infrastructure and procurement of required devices.

6.2 Indicative software implementation estimate

Cost component	Estimated cost (PKR lakh)
Discovery and hospital workflow analysis	15

Cost component	Estimated cost (PKR lakh)
UI/UX and patient/doctor/admin interfaces	25
Backend, database and APIs	45
Mobile/web development	35
AI/NLP and waiting prediction	30
HIS/billing/pharmacy integration	25
Security, RBAC and audit logs	15
Testing and QA	12
Deployment and DevOps	10
Training and pilot support	10
Estimated direct project cost	**222**
Contingency (10%)	22.2
Estimated internal project cost	**244.2**
Indicative contract value including 20% margin on internal cost	**293.0**

Indicative implementation price: PKR 2.9 crore. This is a planning estimate for the defined pilot and first production deployment, not a final quotation.

6.3 Recurring support and payment schedule

Item	Indicative commercial term
Annual maintenance and support (AMC)	12–15% of implementation contract; approximately PKR 40–45 lakh per year for the indicative contract value
AMC scope	Bug fixes, security patches, monitoring, backups, database/application maintenance, integration support, minor improvements, service desk and model monitoring
Out of AMC scope	Major new modules, hospital-wide expansion, new third-party integrations, hardware replacement, paid licenses, major redesigns and change requests
Payment milestone 1	20% at contract signing
Payment milestone 2	20% after architecture and UI approval
Payment milestone 3	25% after core system development demonstration
Payment milestone 4	20% after pilot deployment
Payment milestone 5	10% after UAT/testing acceptance
Payment milestone 6	5% after final go-live and handover

6.4 Assumptions and exclusions

- Hardware is quoted separately after site survey: biometric devices, vitals kiosks/devices, queue displays, tablets/workstations, printers, networking and power backup.
- Third-party license, cloud, SMS/WhatsApp, payment-gateway and device-vendor fees are excluded unless expressly included in the final commercial schedule.
- Final integration cost depends on JPMC HIS and vendor API availability, data quality, interface documentation and security approval.
- Full hospital rollout is a subsequent phase priced after pilot outcome, department count, user volume and integration complexity are confirmed.
- Hospital clinical policy, triage thresholds, prescription authority, patient consent/notice and data retention remain under JPMC governance.

7. Maintenance, Monitoring and Scale-up

7.1 Post-go-live support model

Period	Support approach
Hypercare: first 2–4 weeks	Daily monitoring and review; rapid defect triage; on-site/remote support; user adoption assistance; workflow refinements under change control
Stabilization: months 2–3	Weekly operational review, performance tuning, training refreshers, backlog prioritization and DR/backup verification
AMC: ongoing	Service desk, incident management, patching, monitoring, backup review, integration support, periodic security review and quarterly service reports

7.2 Monitoring and maintenance activities

- Monitor uptime, response times, queue update latency, API errors, failed notifications, integration queues, database capacity and backup status.
- Review access logs, failed login attempts, role changes, sensitive-record access and unusual administrative actions. Security logging should record relevant events without storing credentials or unnecessary sensitive content.
- Patch operating systems, application dependencies and libraries under tested change-control procedures.
- Test backup restoration and disaster-recovery procedures at agreed intervals.
- Monitor AI summary quality, wait-time prediction error, vital-alert rates, user overrides, feedback and potential model drift.
- Maintain a prioritized change backlog; any clinical workflow, data model, integration or AI-behavior change requires approval and regression testing.

7.3 Pilot evaluation and scale-up decision

Dimension	Evaluation method
Operational impact	Compare baseline vs. pilot registration time, queue wait, triage-to-treatment time, doctor utilization and throughput
Patient experience	Survey satisfaction, queue-information usefulness, successful completion and complaint patterns
Clinical experience	Survey doctor/nurse acceptance, summary usefulness, missing-information rate and safety concerns
Technology reliability	Uptime, performance, defect severity, integration success rate, recovery test and support tickets
Governance	RBAC/audit evidence, training completion, incident record and data-quality exceptions
Commercial decision	Assess actual operating cost, hardware needs, integration effort and benefits before approving scale-up

A hospital-wide rollout should be approved only after the pilot acceptance report, corrected priority defects, operational KPI review, security/governance sign-off and an updated commercial scope are completed.

8. Risk Register

Risk	Mitigation
Integration delay or unavailable APIs	Early technical discovery, documented interface dependencies, sandbox access, staged adapters and manual fallback
Poor source data quality	Data profiling, validation, reconciliation, data-owner assignment and scoped migration
Low staff adoption	Co-design workshops, role-based training, champions, beta feedback and simple fallback procedures
Clinical safety concerns	Clinician-led workflow validation, advisory-only AI, auditability, escalation paths and no autonomous diagnosis/prescription
Network/power interruption	Site readiness review, monitored connectivity, backup power, offline/downtime procedure and recovery testing
Cybersecurity incident	Least-privilege RBAC, MFA where feasible, encryption, logging, patching, backups, incident response and regular review
Scope expansion	Change-control board, written impact assessment, phased rollout and separate commercial approval

9. Approval and Next Steps

Upon approval in principle, the project begins with a paid discovery and site-survey phase. That phase confirms the pilot departments, workflow maps, integration availability, hardware list, hosting model, privacy/security requirements, final timeline and binding commercial quotation.

Approval role	Name / signature	Date
JPMC Executive Sponsor		
JPMC Clinical Lead		
JPMC IT/HIS Lead		
Implementation Partner Project Manager		

Appendix: Reference Standards

- HL7 FHIR: use as the interoperability mapping target where appropriate for electronic exchange of healthcare information.
- OWASP ASVS / OWASP security guidance: use as a security requirements and verification reference for the web applications.
- WHO digital-health monitoring and evaluation guidance: use to structure implementation fidelity and outcome measurement.